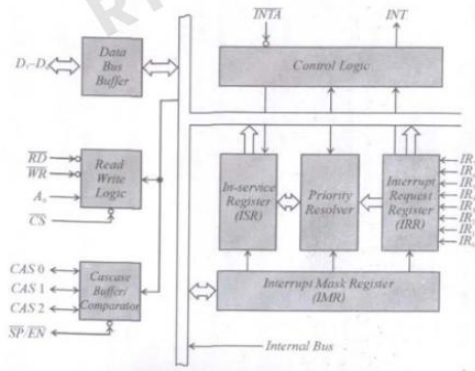
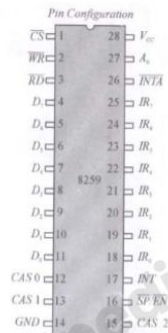


MPI Unit 4 and 5 Quick Revision

8259- PIC(Programmable interrupt controller) 28 pin

PIN DIAGRAM & BLOCK DIAGRAM OF 8259

It is 28 pin IC chip, whose pin diagram and block diagram are as shown:



8 additional interrupt - IN 0- TO IN7

8 line minimum and by cascading 8x8 lines

registers- 3 ISR, IRR, IMR

SP/EN -> master -slave interrupt

data bus buffers- triastate (high impedance for 1 , logic 0 and logic 1 for low)

Interfacing in 8085 Microprocessor

Memory Organization of 8085

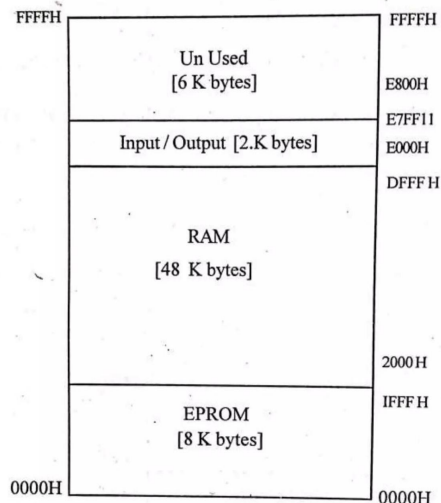


Fig. 3.1 Address map of 64 K bytes Address Space

Memory Mapped I/O

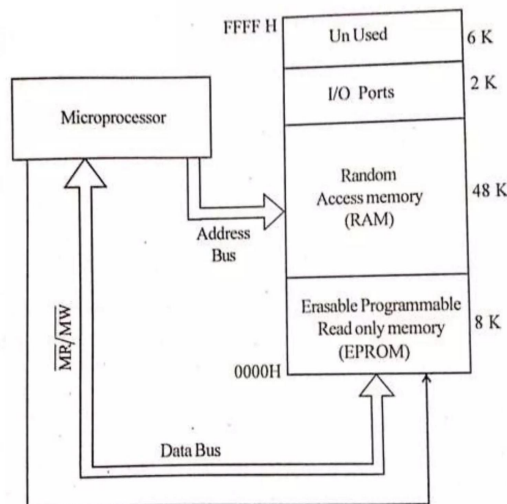
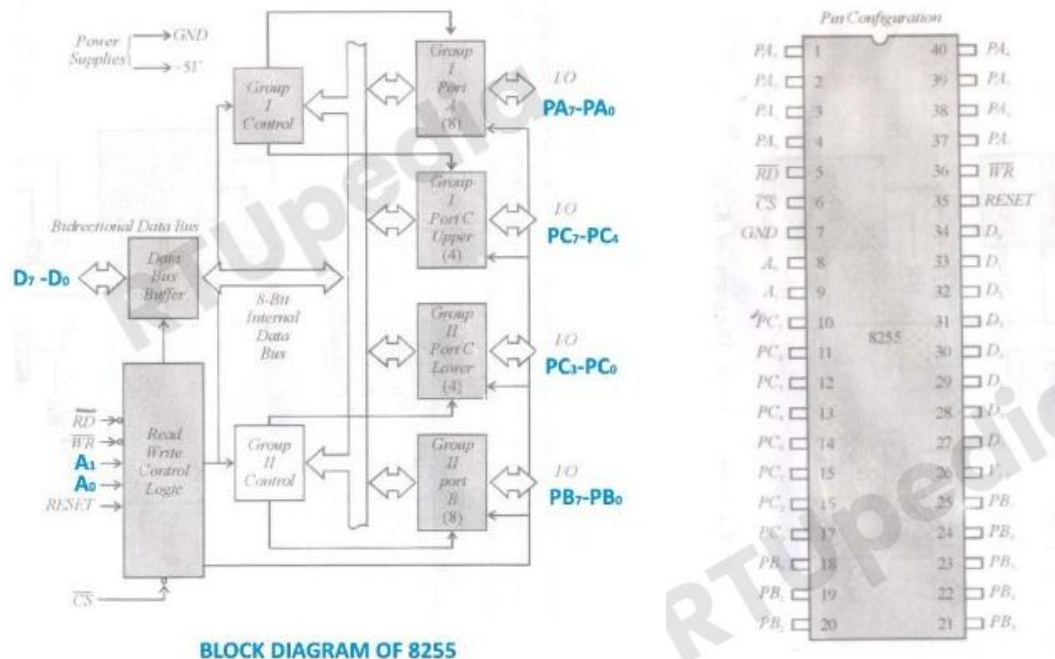


Fig.3.4 Memory mapped I/o (64 Kbyte address space)

connect processor to peripheral device or memory to communicate ->Interfacing component-> tristate buffer (increase capability by power increment),
encoder (generate code for one of the input signal) ,
decoder(identifies the combination and signals),
Latch (hold data to display data on output)

8255 - PPI (Programmable Peripheral interfacing) -



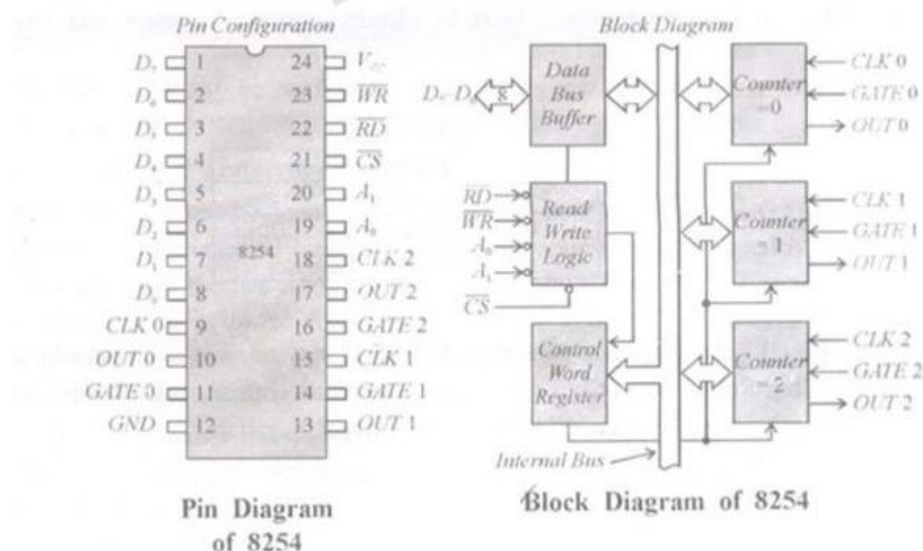
> parallel output

8 port -> A B and C with

40 pin PA, PB, PC -> $8 \times 3 + 8$ data pin = 32 then other pins for write read vcc and gnd
internal register- control reg. defines modes of 8255

8254/8253 (PIT) Programmable interval timer

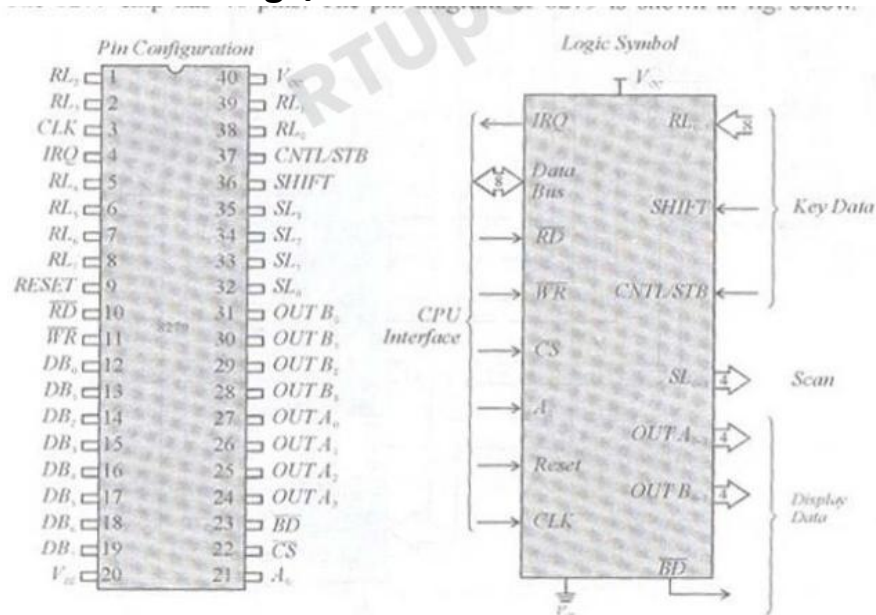
The 8254 chip has 24 pins. The pin diagram is shown in fig.



(24 pin) provide accurate time delay

16 bit counter- counter 0, 1 and 2 -> clk 0, 1 and 2 and gate 0, 1 and 2 with out 0, 1 and 2

8279 - Interfacing I/O device



FIFO principle

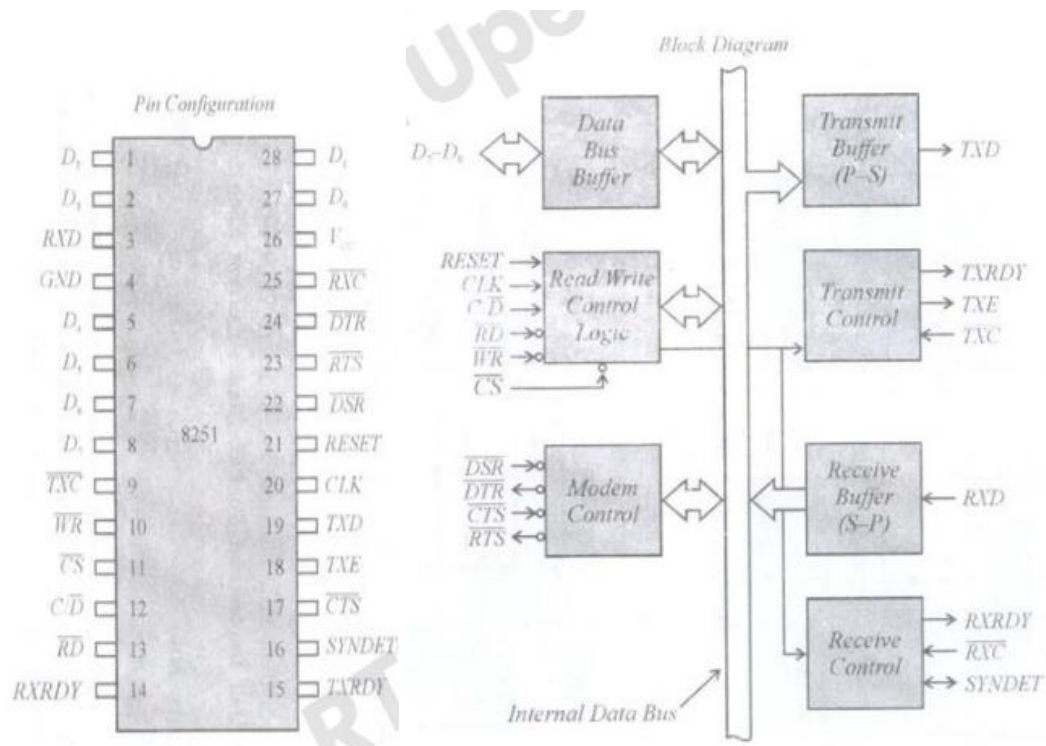
keyboard / display controller

40 pin output line- outA 0 to 3 and outB 0 to 3

Shift SL 4 , IRQ and 4 RL pins

Programmable Communication interface - 8251 (USART)

Universal Sync and Async receiver and transmittor - 28 pins



Difference Between RS-232 And RS-422

Feature	RS-232	RS-422
Type of signaling	Single-ended	Differential
Voltage levels	$\pm 3V$ to $\pm 15V$	$\pm 2V$ (difference between lines)
Noise immunity	Low	High
Data transmission	Short distance	Long distance
Maximum distance	~15 meters	~1200 meters
Speed	Low (up to ~20 kbps typical)	High (up to ~10 Mbps)
Number of devices	Point-to-point (1 transmitter, 1 receiver)	Multi-point (1 driver, multiple receivers)
Wiring	Single wire + ground	Twisted pair (2 wires per signal)
Cost/complexity	Simple, low cost	More complex

IEEE-488

IEEE-488, also known as **GPIB (General Purpose Interface Bus)**, is a standard used for communication between computers and electronic instruments.

IEEE-488 uses **24 lines**, divided into:

- **8 Data lines (DIO1–DIO8)**
- **3 Handshake lines**
- **5 Management/control lines**
- **8 Ground lines**

Key Features

- Parallel data transfer (8-bit data bus)
- Supports **multiple devices (up to 15)**
- Maximum cable length $\approx$ **20 meters**
- High speed and reliable communication

